

Oscillations

Basic skills

Q1. $x(t) = A \cos \omega t + B \sin \omega t$

Q2. Yes only tension $\rightarrow +ve$
 $F = -T$
 $Ma = -kx$
Centre

So $\ddot{x} = -\frac{k}{m}x$ $\omega = \sqrt{\frac{k}{m}}$

Q3. $\ddot{x} + k\dot{x} + 9x = 0$

a) Light if slow decay oscillation

so $d < 0$ so $k^2 - 4 \times 9 < 0$

$k^2 < 36$ $k < 6$

$\& k > 0$ as well

b) Heavy if sharp so $d > 0$

so $k^2 - 36 > 0$ so $k > 6$

c) critical if between so $d = 0$

$k^2 - 36 = 0$ $k = 6$

Q4. $f = \frac{1}{P}$ so $P = \frac{1}{f} = \frac{1}{740}$ seconds
 AS Hz is per sec

Competency skills

Q5. $f = 1800 \text{ Hz}$, $A = 2 \text{ mm}$

a) As $f = \frac{\omega}{2\pi} \Rightarrow \omega = f \times 2\pi$
 $= 3600\pi$

so $\ddot{x} = -(3600\pi)^2 x$

b) $v^2 = \omega^2 (A^2 - x^2)$ max at $x = 0$

for $v^2 = \omega^2 A^2$ so $v = \omega A$

so $v = 3600\pi \times 2 = 7200\pi \text{ mm s}^{-1}$
 $= 7.2\pi \text{ m s}^{-1}$ or $2 \text{ mm} = 0.002 \text{ m}$

For acceleration $\ddot{x} = -(3600\pi)^2 x$

so max at $x = A$ so $\ddot{x} = -(3600\pi)^2 \times 0.002$
 $= -25920\pi^2$

so $25920\pi^2$

c) At $x = 0.001\text{m}$

$$v^2 = (3600\pi)^2 (0.002^2 - 0.001^2)$$

$$v = 3600\pi \sqrt{0.002^2 - 0.001^2}$$

$$= 19.6\text{ m s}^{-1}$$

$$\ddot{x} = - (3600\pi)^2 \times 0.001$$

$$= -12960\pi^2$$

Q6.

As $x = R \cos(\omega t + \phi)$

for R, ϕ but Amplitude A and Phase difference only due to initial position so ignore ($\phi = 0$)

$$x = A \cos(\omega t)$$

$$\dot{x} = -\omega A \sin(\omega t)$$

Now $\sin(\omega t) = \sqrt{1 - \cos^2 \omega t}$

so $\dot{x} = -\omega A \sqrt{1 - \cos^2 \omega t}$

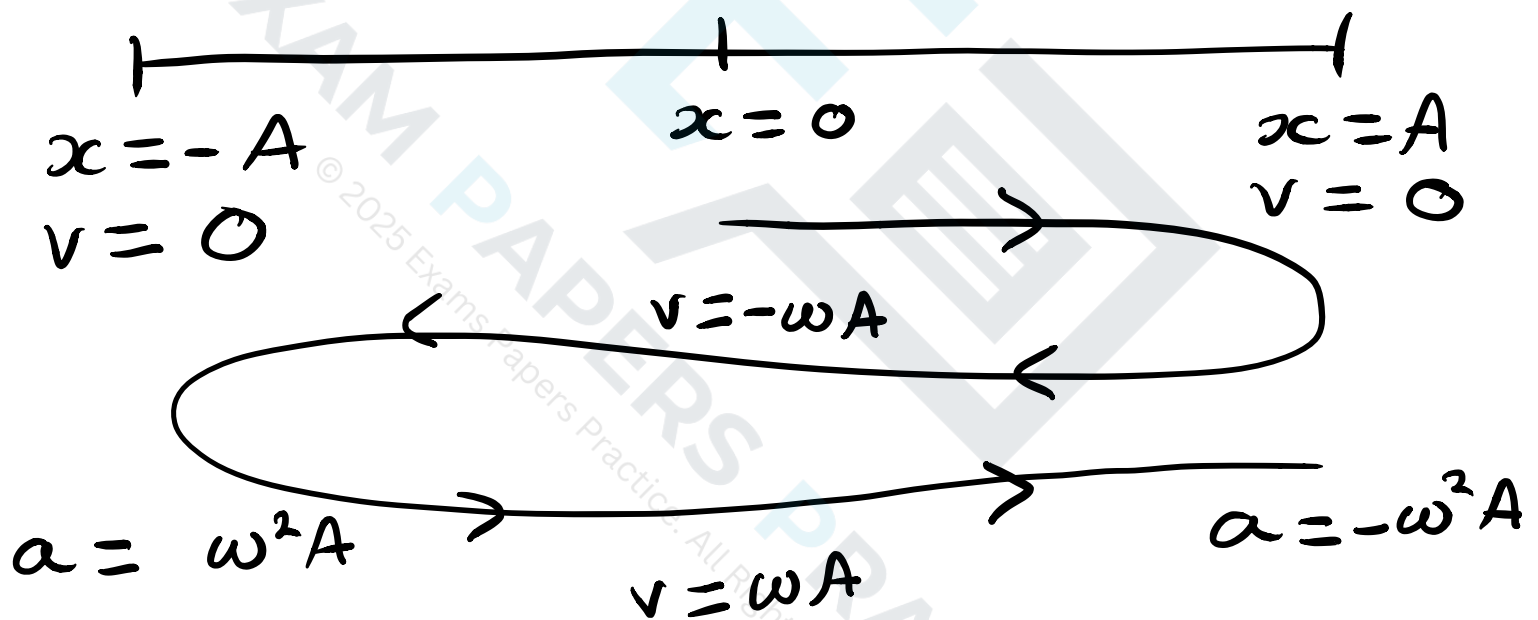
$$= -\omega A \sqrt{1 - \left(\frac{x}{A}\right)^2}$$

AS $\dot{x} = v$

$$v = -\omega \sqrt{A^2 - x^2}$$

so $v^2 = \omega^2 (A^2 - x^2)$

Q7.



Q8.

so 2000 rpm is $2000 \times 60 = 120000$ per second $f = 120 \text{ kHz}$

so $\omega = 2\pi f = 240\pi \times 10^3$

so $v = \omega \sqrt{A^2 - x^2} = (240\pi \times 10^3) \sqrt{0.2^2 - 0.01^2}$

$\Rightarrow v \approx 150607 \text{ m s}^{-1}$

Q9.

- a) B as it's got the highest amplitude
- b) C as it's got shortest period
- c) NO waveform seems to decay so little to no damping